ELEMENT D: WEIGHTED DECISION MATRIX (PUGH MATRIX)
Instructions
1. List measurable design requirements (criteria)
2. Assign weights (1-5) for importance (RED NUMBERS)
3. Score ideas (1-10) based on percieved performance (Informed Guessing)
4. Calculate weighted values (Weight X Score)
5. Highest score is the data-driven choice to move forward

SCORING KEY (1-10)
9–10: Perfect solution; exceeds the requirement.
7–8: Strong solution; meets all aspects of the requirement.
4–6: Functional; meets the requirement but with known trade-offs.
1–3: Poor; barely meets the requirement or creates new problem

		dea #1: Basic Table w/ Brac		Idea #2: 4legs on storage		Idea #3: Puzzle Piece table		4: Triangle from ground to		Idea #5: Skirt table	
Design Requirements	**Weight (1-5)**	**Score (1-10)**	**Weighted Value**	**Score (1-10)**	**Weighted Value**	**Score (1-10)**	**Weighted Value**	**Score (1-10)**	**Weighted Value**	**Score (1-10)**	**Weighted Value**
It must support Cooper a	5	8	40	10	50	5	25	7	35	8	40
The design is replicable,	4	10	40	8	32	6	24	8	32	4	16
When applied with a rep	3	8	24	9	27	4	12	8	24	7	21
It must have enough clea	2	9	18	7	14	8	16	8	16	10	20
It must be easily fixable.	1	9	9	7	7	1	1	7	7	4	4
Total Weighted Score		44	**131**	41	130	24	78	38	114	33	101

Winning Rational Write a concise justification of your top scoring design based on your design requirements	Idea #2: A basic table wth bracing will be extremly sturdy (as seen from litterarly every table in existance) will be both extremly easy for Mr.L to understand and reproduce himself. Furthermore it will be able to be consstantly replicated. Along with this there will be nothing in the wa for a chiar besides the legs, whcih is why it outscoured the storage one, as the stroage could potentially get in the way of students. Furthermore to fix this design there would be plenty of room under the table to fix it, unlike the other desings, and all it SHOULD take is some screws and bolts being tightened

Design Requirement 1	It must support Cooper and Abel's full weight without giving, meaning the braces of the legs don't wobble/move
Design Requirement 2	The design is replicable, can Mr.L read through the instructions/diagrams we provide him without any questions
Design Requirement 3	When applied with a repeated amount of large force (someone hitting it with a hammer/mallet, jumping on it, something along those lines) x amount of times to the table, the legs don't move more than 1cm.
Design Requirement 4	It must have enough clearance to fit a chair base fully under it (pushed in)
Design Requirement 5	It must be easily fixable. If screws are loosened/leg falls off, it must take less than 1 minute to put back on.

Idea 1	It is a 4 legged table with triagular bracing between legs, with a metal bracing under the table top and in between the legs.
Idea 2	It is a 4 legged design that sits on a a metal storage base, acting as a 5th leg.
Idea 3	It is a table that connects via wood pieces that enables us to not use bracing.
Idea 4	Wooden beems would connect the bottom of the table corners to the middle of the table.
Idea 5	The middle of the table would have 3 cut beams that would extend to the floor, making a pyrimidal shape.